

A tree diagram

A tree diagram represents a sequence of outcomes with branches.

What two first branches represent one fair-coin toss?

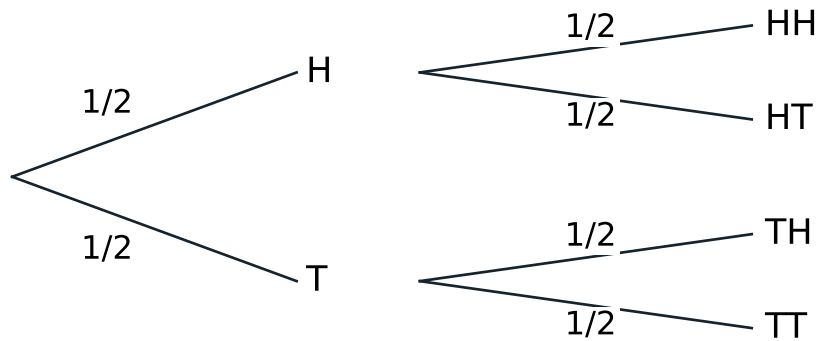
Build a two-stage tree

Toss a fair coin twice independently.

How would you label every branch and terminal outcome?

Multiply along a path

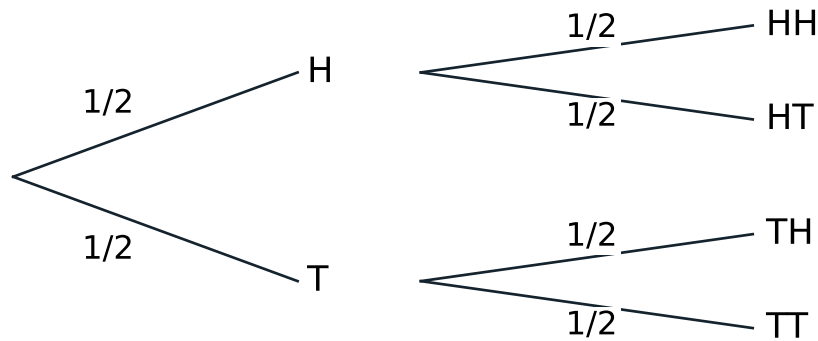
Two independent fair-coin tosses.



What is the probability of the H then T path?

Add terminal paths

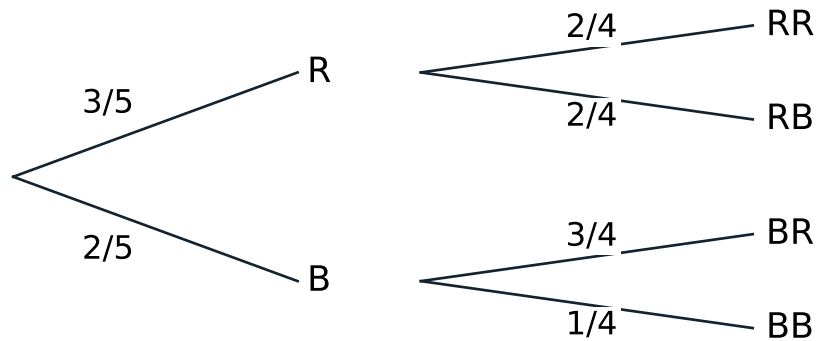
Two independent fair-coin tosses.



Which paths contribute to exactly one head?

Tree without replacement

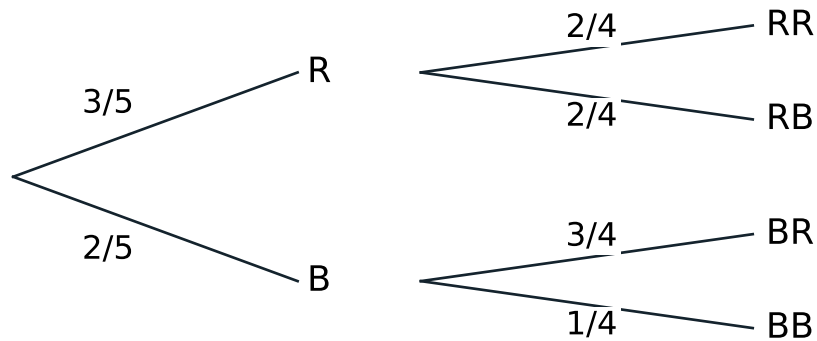
3 red, 2 blue; uniform draws, no replacement.



Why is $P(\text{red second} \mid \text{red first}) = 2/4$?

A second set of branches

3 red, 2 blue; uniform draws, no replacement.



What is the probability of the B then R path?

Branch totals

A tree node has outgoing probabilities 0.3, 0.4, and 0.4 for mutually exclusive exhaustive outcomes.

What is wrong with this node?

A tree with replacement

A box holds 3 red and 2 blue tokens. Replace and mix after each uniform draw.

How do the second-stage branch labels differ from drawing without replacement?

Frequency-labeled trees

A box has 2 labeled red and 3 labeled blue tokens. Draw twice with replacement and mixing.

How many ordered token pairs follow each color path, and how many exist altogether?

A weighted path

Synthetic model: 40% of jobs use server A, 60% use B; completion probabilities are 0.9 on A and 0.8 on B.

What is the probability of choosing A and completing the job?

Sum successful paths

Synthetic model: 40% of jobs use server A, 60% use B; completion probabilities are 0.9 on A and 0.8 on B.

What is the overall probability of completion?

Reverse the conditioning

Synthetic model: 40% of jobs use server A, 60% use B; completion probabilities are 0.9 on A and 0.8 on B.

Given completion, what is the probability the job used A?

A Venn diagram

A Venn diagram places overlapping event regions inside the sample space.

Where would an outcome belonging to both A and B appear?

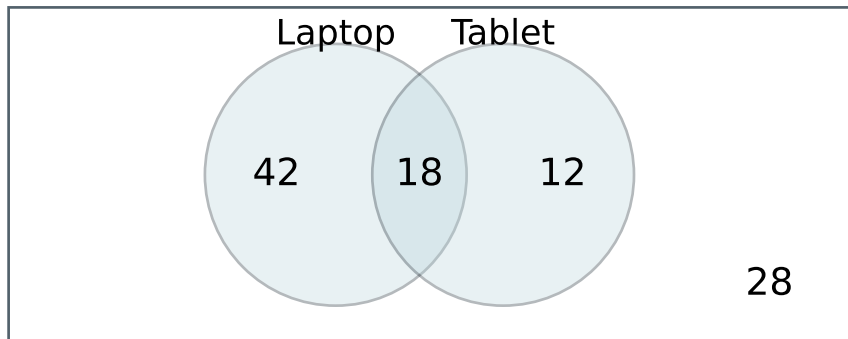
Fill the regions

Synthetic survey of 100 students: 60 use laptops, 30 use tablets, and 18 use both.

What counts belong in the four regions of a two-circle Venn diagram?

Only one category

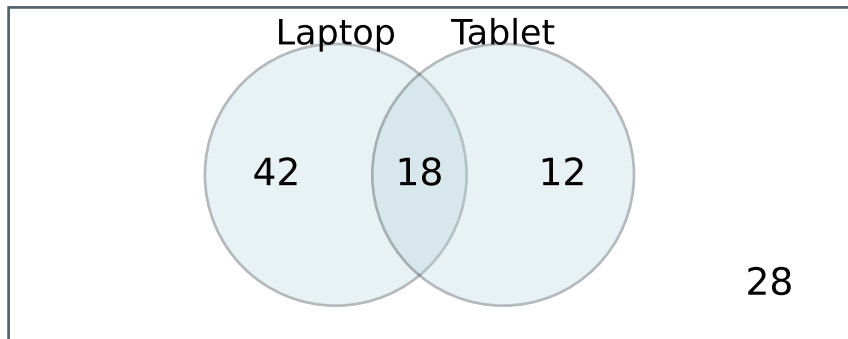
Synthetic survey of 100 students.



How many use a laptop but no tablet?

Outside both circles

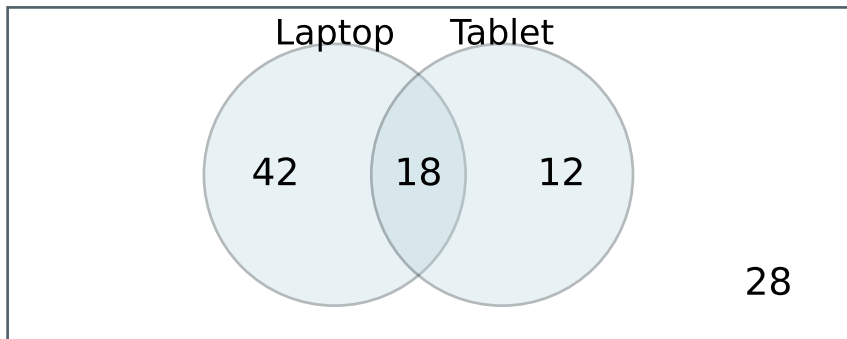
Synthetic survey of 100 students.



How many use neither device?

Shade a union

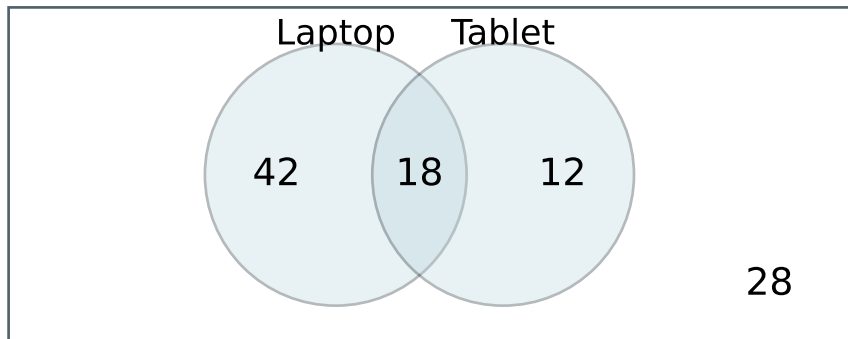
Synthetic survey of 100 students.



Which regions form laptop OR tablet?

Shade a complement

Synthetic survey of 100 students.



Which region complements laptop OR tablet?

Conditional probability in a Venn diagram

Synthetic model: $P(A) = 0.6$, $P(B) = 0.4$, $P(A \text{ AND } B) = 0.2$.

After drawing the Venn regions, what is $P(A | B)$?

Disjoint circles

Two positive-probability events have no overlap in a Venn diagram.

What does this show, and does it establish independence?